LAB- 11

WAP in 8086 to take a string from user and print that string in uppercase or in lowercase.

PROGRAM:

```
.MODEL SMALL
.STACK 100H
.DATA
STR1 DB 'ENTER A STRING:','$'
STR2 DB 30 DUP ('$')
.CODE
  MAIN PROC
    MOV AX, @DATA
    MOV DS, AX
    LEA DX, STR1                ; LOAD ADDRESS OF STR1 IN DX
    MOV AH, 09H                 ; DISPLAY STRING WITH ADDRESS IN DX
    INT 21H
    LEA DX, STR2                ; LOAD ADDRESS OF STR2 IN DX
    MOV AH, 0AH                 ; STORE INPUT STRING IN ADDRESS IN DX
    INT 21H
    LEA SI, STR2                ; LOAD ADDRESS OF STR2 IN SI REG.
    MOV CX, [SI+1]              ; LOAD ADDRESS OF STR2 IN CX
    MOV CH, 00H                 ; CH=00H
PRINT: MOV DL, [SI+2]           ;LOAD FIRST CHARACTER OF STRING IN DL
    SUB DL, 20H                 ; DL = DL-20H (LOWERCASE TO UPPERCASE
    MOV AH, 02H                 ; DISPLAY CHARACTER STORED IN DL REG
    INT 21H
    INC SI                      ; NEXT CHARACTER OF STRING IN SI
    LOOP PRINT
    MOV AH, 4CH
    INT 21H
    MAIN ENDP
  END MAIN
```

Write a Program to reverse the string enter by user.

```
org 100h
.model small
```

```
.stack 100h
   .data
 .code
 main proc
   mov cx,0 ;set counter
    j: mov ah,01
   int 21h
   mov bl,al
   inc cx
   cmp bl,0Dh; check whether the user provided the input by pressing enter key
                              ; ODh is ASCII key for enter key
   je rev    ; if equl then jump on rev; where rev is used to print  reverse string
   push bx   ; if not equal then push bx register.
   jmp j    ; execute till user not pressed enter key.
   rev:
   mov dl, 0Ah ;
   mov ah, 02  ;
   int 21h     ; these entire code is for new line print
   mov dl,0Dh  ;
   mov ah,02   ;
   int 21h     ;
  l: pop dx
   mov ah,02
   int 21h
   loop l
  main endp
end main
ret
```

write a Program to reverse string.

```
Data Segment
  str1 db 'HELLO WORLD','$'
  strlen1 dw $-str1
  strrev db 20 dup(' ')
Data Ends
Code Segment
  Assume cs:code, ds:data
  Begin:
    mov ax, data
    mov ds, ax
    mov es, ax
    mov cx, strlen1
    add cx, -2
    lea si, str1
    lea di, strrev
    add si, strlen1
    add si, -2
    L1:
      mov al, [si]
      mov [di], al
      dec si
      inc di
      loop L1
      mov al, [si]
      mov [di], al
      inc di
      mov dl, '$'
      mov [di], dl
    Print:
      mov ah, 09h
```

```
    lea dx, strrev
    int 21h
  Exit:
    mov ax, 4c00h
    int 21h
Code Ends
End Begin
```